MAT 201 - MATHEMATICAL METHODS I

LECTURE 8: INCREMENTS, DIFFERENTIALS AND LINEAR APPROXIMATIONS

LECTURE C24-1-8 :: MON., FEB. 03, 2025 8:00 AM (WAT) - FRI., FEB. 07, 2025 6:00 PM (WAT)

Prepared by Dr. Ashiribo Senapon WUSU
Email: ashiribo.wusu@lasu.edu.ng | Phone: 08026479471

Mathematics, Lagos State University.

OUTLINE

- Introduction
- Increments & Differentials
- Linear Approximations & Applications
- Assignment

0:00

INTRODUCTION

0:00

INCREMENT

Let $f: X \subseteq \mathrm{R}^n \to \mathrm{R}$ and let $\mathbf{a} \in X$, the incremental change of f denoted as $\triangle f$, is

$$\triangle f = f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a})$$

0:00

TOTAL DIFFERENTIAL

The total differential of f, denoted as $df(\mathbf{a}, \mathbf{h})$, is

$$df(\mathbf{a}, \mathbf{h}) = \frac{\partial f}{\partial x_1}(\mathbf{a})h_1 + \frac{\partial f}{\partial x_2}(\mathbf{a})h_2 + \cdots + \frac{\partial f}{\partial x_n}(\mathbf{a})h_n$$

The significance of the differential is that for $h \approx 0$,

$$\triangle f = df$$

0:00

TOTAL DIFFERENTIAL

Sometimes, h_i is replaced by the expression $\triangle x_i$ or dx_i to emphasize that it represents a change in the ith independent variable, in which case we write

$$\begin{aligned} df = & \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \cdots + \frac{\partial f}{\partial x_n} dx_n \\ = & \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} dx_j \end{aligned}$$

REMARKS: We've suppressed the evaluation of the partial derivatives at $\mathbf{a}$, as is customary.

EXAMPLE 1

Suppose $f(x, y, z) = \sin(xyz) + \cos(xyz)$, then

$$\begin{aligned} df = & \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \cdots + \frac{\partial f}{\partial z}dz + \\ = & yz[\cos(xyz) - \sin(xyz)]dx + xz[\cos(xyz) - \sin(xyz)]dy + \\ & xy[\cos(xyz) - \sin(xyz)]dz \\ = & [\cos(xyz) - \sin(xyz)](yzdx + xzdy + xydz) \end{aligned}$$

0:00

EXAMPLE 2

Let $f(x, y) = x - y + 2x^2 + xy^2$, then for $(a_1, a_2) = (2, -1)$, we have that the increment is

$$\begin{aligned}\triangle f &= f(2 + \triangle x, -1 + \triangle y) - f(2, -1)\\ &= 2 + \triangle x - (-1 + \triangle y) + 2(2 + \triangle x)^2 +\\ &\qquad (2 + \triangle x)(-1 + \triangle y)^2 - 13\\ &= 10\triangle x - 5\triangle y + 2(\triangle x)^2 - 2\triangle x\triangle y + 2(\triangle y)^2 + \triangle x(\triangle y)^2\end{aligned}$$

EXAMPLE 2 [CONT.]

On the other hand,

$$\begin{aligned} & df((2,\ -1), (\triangle x, \triangle y)) \\ = \ & f_x(2,\ -1)\triangle x + f_y(2,\ -1)\triangle y \\ = \ & \left[\left(1 + 4x + y^2\right)\right]_{(2,-1)}\triangle x + [(-1 + 2xy)]_{(2,-1)}\triangle y \\ = \ & 10\triangle x - 5\triangle y \end{aligned}$$

EXAMPLE 2 [CONT.]

We see that df consists of exactly the terms of $\triangle f$ that are linear in $\triangle x$ and $\triangle y$ (i.e., appear to first power only). This will always be the case, of course, since that is the nature of the first-order Taylor approximation. Use of the differential approximation is often sufficient in practice, for when $\triangle x$ and $\triangle y$ are small, higher powers of them will be small enough to make virtually negligible contributions to $\triangle f$.

0:00

EXAMPLE 3

A wooden rectangular block is to be manufactured with dimensions $3in \times 4in \times 6in$. Suppose that the possible error in measuring each dimension of the block is the same. We use differentials to estimate how accurately we must measure the dimensions so that the resulting calculated error in volume is no more than $0.1in^3$.

0:00

EXAMPLE 3 [CONT.]

Let the dimensions of the block be denoted by $x(\approx 3in)$, $y(\approx 4in)$, and $z(\approx 6in)$. Then the volume, V of the block is:

$$V = xyz \quad \text{and} \quad V \approx 3in \times 4in \times 6in = 72in^3$$

The error in calculated volume is $\triangle V$, which is approximated by the total differential dV. Thus

$$\begin{aligned} \triangle V \approx dV &= V_x(3, 4, 6)\triangle x + V_y(3, 4, 6)\triangle y + V_z(3, 4, 6)\triangle z \\ &= 24\triangle x + 18\triangle y + 12\triangle z \end{aligned}$$

EXAMPLE 3 [CONT.]

If the error in measuring each dimension is ϵ, then we have

$$\triangle x = \triangle y = \triangle z = \epsilon$$

Therefore,

$$dV = 24\triangle x + 18\triangle y + 12\triangle z = 24\epsilon + 18\epsilon + 12\epsilon = 54\epsilon$$

To ensure (approximately) that $|\triangle V| \leq 0.1$, we demand $|\triangle V| = |54\epsilon| \leq 0.1$. Hence, $|\epsilon| \leq \frac{0.1}{54} = 0.0019in$. So, the measurements in each of the dimensions must be accurate to within $0.0019in$.

EXAMPLE 4

The formula for the volume of a cylinder of radius r, and height h, is $V(r, h) = \pi r^2 h$. If the dimensions are changed by small amounts $\triangle r$ and $\triangle h$, then the resulting change $\triangle V$ in volume is a approximated by the differential change dV. That is

$$\triangle V \approx dV = \frac{\partial V}{\partial r}\triangle r + \frac{\partial V}{\partial h}\triangle h = 2\pi r h \triangle r + \pi r^2 \triangle h$$

0:00

EXAMPLE 4 [CONT.]

Suppose the cylinder is actually a softdrink can, so that it has approximately dimensions of $r = 1in$ and $h = 5in$. Then

$$dV = \pi(10\triangle r + \triangle h)$$

EXAMPLE 4 [CONT.]

This statement shows that, for these particular values of r and h, the volume is approximately 10 times more sensitive to change in radius than change in height. i.e., if the radius is changed by an amount ϵ, then the height must be changed by roughly 10ϵ to keep the volume constant (i.e., to make $\triangle V$ zero).
We used the word "approximate" because our analysis arises from considering the differential change dV, rather than the actual incremental change $\triangle V$.
This softdrink can example has real application to product marketing strategies. Because the volume is so much sensistive to change in radius than height, it is possible to make a can appear to be larger than standard by decreasing its radius slightly and increasing the height so no change in volume occurs.

LASU-VLAP

You have to complete this lecture within Mon., Feb. 03, 2025 8:00 AM (WAT) - Fri., Feb. 07, 2025 6:00 PM (WAT) in order to mark attendance.

OK